Do LLM-Judge Biases Compose? A Label-Free, Scale-Aware Test for Position–Length Interaction

Anonymous authors
Paper under double-blind review

Abstract

Pipelines that use large language models as pairwise judges remove position bias and length bias in separate steps, assuming the two biases add. We show that this assumption matters, can be tested without human labels, and is easy to test wrongly. On the win-rate scale, where leaderboards and preference labels live, we prove that *any* correction applying one position offset across length conditions misstates the verbose-versus-terse contrast by exactly half the position×length interaction, for every judge and every link function; swap-averaging and length-stratified correction are exact. The Swap-Symmetry Interaction Test (SSIT) estimates this interaction from the four position×length swap cells of each stimulus, as a difference of cell means that needs no labels and no model of the judge. Two artefacts can manufacture the very effect SSIT measures. Coding a tie as a vote for the second slot, as our own first analysis did, breaks swap symmetry; we prove that it adds the fall in tie rate under lengthening to the estimate, and that it makes a position-invariant judge look position-biased. Near the ceiling, link curvature turns additive log-odds biases into win-rate interaction; we give a per-judge curvature baseline. Re-analysing seven judges from six model families shows how much this matters: slot-coded, six judges appear to interact; with swap-equivariant coding and inference clustered on the 12 distinct stimuli, none does individually. A pooled effect survives: +4.7 win-rate points (95% CI $[1.6, 7.8]$), above the curvature baseline, with log-odds interaction 0.67 $[0.35, 1.18]$. Position bias is absent when lengths match (-0.1 points) and appears only when the target is longer (+4.6). We release code, data, and a protocol sized for per-judge conclusions.

1 Introduction

LLM judges now rank models on public leaderboards and label the preference data used to train them (Zheng et al., 2023; Dubois et al., 2023; Chiang et al., 2024; Lee et al., 2023). Their systematic biases are well catalogued: a judge's verdict depends on which slot a response occupies (Wang et al., 2024; Shi et al., 2024), on its length (Saito et al., 2023; Dubois et al., 2024; Singhal et al., 2023), on its style (Wu & Aji, 2023), and on whether the judge wrote it (Panickssery et al., 2024; Wataoka et al., 2024); surveys and audits organize these one bias at a time (Ye et al., 2025; Gu et al., 2024; Li et al., 2024; Thakur et al., 2024; Koo et al., 2023). Mitigations follow the same convention. Position bias is removed by swapping slots or by subtracting an estimated offset; length bias by regression on a length covariate (Dubois et al., 2024). A deployed pipeline chains these steps.

Chaining is valid only if the biases *compose additively*: the position offset must not depend on the length gap, and the reverse. If it does, a pipeline can be correct for each bias in isolation and still wrong for their combination, and the error lands precisely on the comparisons length control was meant to repair. The natural check compares the judge with human preferences, which defeats the purpose of a judge. This paper asks whether the composition assumption can be checked *without labels*, and what it takes to check it correctly.

Contributions.

1. **Exposure, exactly (§3).** On the win-rate scale, any correction that applies one position offset across length strata, however it is estimated, misstates the verbose-versus-terse contrast by exactly $\frac{1}{2}\iota_{PV}$, where ι_{PV} is the position×length interaction in win rate (Proposition 1). The result holds for every judge, because it concerns cell means only. Swap-averaging and length-stratified correction are exact. We explain why the tempting log-odds version of this statement fails through non-collapsibility.
2. **A label-free test (§4).** SSIT presents each stimulus in the four position×length swap cells and averages a per-stimulus difference-in-differences. It is unbiased for ι_{PV} by design (Proposition 2). We pair it with a sign-flip randomization test, wild-cluster intervals, and a sample-size rule driven by quantities observed in a short pre-run.
3. **Two artefacts, with remedies (§5).** (a) *Tie coding.* Counting a tie as a vote for slot B breaks the swap symmetry the design depends on. It biases the interaction by the fall in tie rate under lengthening (Proposition 3) and makes a position-invariant judge appear biased toward slot B (Corollary 2). (b) *Curvature.* A judge with additive log-odds biases still shows win-rate interaction near the ceiling. That interaction requires a position main effect (Proposition 4), and we estimate it per judge.
4. **A corrected multi-family pilot (§7).** On seven judges from six families, the two artefacts account for most of what slot-coded analysis reports per judge (Figure 1). A modest pooled interaction survives every correction, and it has the structure the theory predicts: no position bias at matched length, a positive one when the target is longer. A length-only manipulation retains about 70% of it but is not significant on its own.

We also give a label-free, single-pass correction (IASC, §6) as a fallback, and a confirmatory protocol (Appendix D). Our pilot data are deliberately small, and we frame them as a demonstration of the diagnostic: they show how far a careful analysis moves the conclusions, and they size the study that would settle per-judge questions.

2 Related work

Judge biases and their mitigation. Position bias was documented when LLM judging was introduced (Zheng et al., 2023) and is mitigated by evaluating both orders (Wang et al., 2024); Shi et al. (2024) study it systematically across judges and tasks. Length bias is documented for preference labelling (Saito et al., 2023) and reward modelling (Singhal et al., 2023). Length-controlled AlpacaEval removes it by regressing win rate on a length term (Dubois et al., 2024), which is the *additive* regime our Proposition 1 concerns. Self-preference (Panickssery et al., 2024; Wataoka et al., 2024) and style (Wu & Aji, 2023) are treated in the same one-bias-at-a-time way. Soumik (2026) compare several debiasing strategies across judges against human labels and find position bias small *in isolation*. Our results are compatible: in our pilot the position main effect is near zero once ties are coded symmetrically, while its interaction with length is not. To our knowledge no prior work tests whether the corrections compose, or does so without labels.

Interaction depends on scale. Whether two factors interact depends on the scale of measurement (VanderWeele & Knol, 2014). In logit and probit models, interaction in coefficients and interaction in probabilities can differ in size and even in sign (Ai & Norton, 2003). Logistic coefficients are also non-collapsible: marginalizing over a covariate changes a coefficient even without confounding (Greenland et al., 1999), and marginal coefficients are attenuated relative to conditional ones under heterogeneous intercepts (Zeger et al., 1988). We use these facts to put the theory on the scale where pipelines report numbers (§3), and to separate interaction that is intrinsic to the judge from interaction produced by curvature (§5.2).

Ties in paired comparison. Paired-comparison models treat ties as a third outcome (Rao & Kupper, 1967; Davidson, 1970). LLM-judge pipelines often collapse ties instead, either dropping them or mapping them to a slot. We show that mapping them to a slot is not innocuous under a swap design.

Label-efficient evaluation. Prediction-powered inference combines a small gold set with many judge labels to give valid intervals (Angelopoulos et al., 2023a;b; Chen et al., 2026). It corrects judge *error* against labels; SSIT tests a structural assumption of the debiasing pipeline without them. The two are complementary (§8).

3 Setup and what an additive correction leaves behind

Design and outcome. A stimulus k consists of a prompt, a *target* response, and a comparison response. A cell $(p, v) \in \{0, 1\}^2$ fixes the slot (p=1: target shown first, as Response A) and the length condition (v=1: the target is presented in a lengthened form). The judge returns A, B, or a tie. Throughout, outcomes are on the scale of *whether the target was chosen*. A coding is *swap-equivariant* if it depends only on whether the verdict names the target, names the comparison, or is a tie, and never on which slot those responses occupy. The primary coding is

$$Y = 1 \text{ (target named)}, \qquad Y = \tfrac{1}{2} \text{ (tie)}, \qquad Y = 0 \text{ (comparison named)}. \tag{1}$$

Let $r_k(p, v) = \mathbb{E}[Y \mid k, p, v]$ be the cell rate. Define the *slot-balanced win rate* $\theta_k(v) = \frac{1}{2}[r_k(1, v) + r_k(0, v)]$, the *position effect* $\Delta_k(v) = r_k(1, v) - r_k(0, v)$, and the *position×length interaction*

$$\imath_k = \Delta_k(1) - \Delta_k(0) = [r_k(1, 1) - r_k(0, 1)] - [r_k(1, 0) - r_k(0, 0)], \qquad \imath_{PV} = \mathbb{E}_k[\imath_k], \tag{2}$$

with the expectation over the stimulus population. $\imath_{PV}$ is in win-rate units. The estimand a debiased pipeline aims at is $\theta(v) = \mathbb{E}_k \theta_k(v)$, together with contrasts such as $\theta(1) - \theta(0)$, the length effect free of position.

Proposition 1 (Additive position corrections, win-rate scale)**.** *Consider a single-order evaluation that shows the target in slot p=1 and corrects the observed rate $r(1, v)$ by subtracting a constant c that does not depend on v. Its error for $\theta(v)$ is $e(v) = \frac{1}{2}\Delta(v) - c$. Hence:*

- *(a) for every c, the error in the length contrast is $e(1) - e(0) = \frac{1}{2}\imath_{PV}$;*
- *(b) $\max_v |e(v)| \geq \frac{1}{4}|\imath_{PV}|$, with equality exactly for the marginal estimate $c = \frac{1}{4}[\Delta(0) + \Delta(1)]$ obtained from a calibration set balanced over v, which leaves $e(1) = +\frac{1}{4}\imath_{PV}$, $e(0) = -\frac{1}{4}\imath_{PV}$;*
- *(c) swap-averaging, which reports $\frac{1}{2}[r(1, v) + r(0, v)]$, and the stratified correction $c_v = \frac{1}{2}\Delta(v)$ are exact for every judge.*

Proof. $r(1, v) = \theta(v) + \frac{1}{2}\Delta(v)$ by definition, so $r(1, v) - c - \theta(v) = \frac{1}{2}\Delta(v) - c$. Part (a) follows by differencing. For (b), $|e(1)| + |e(0)| \geq |e(1) - e(0)| = \frac{1}{2}|\imath_{PV}|$, so the larger term is at least $\frac{1}{4}|\imath_{PV}|$, with equality iff $e(1) = -e(0)$, i.e. $c = \frac{1}{4}[\Delta(0) + \Delta(1)]$. Part (c) is immediate from the definitions. □

The statement concerns cell means only. It makes no assumption about how the judge decides, holds equally for a judge that is a lookup table, and does not depend on the order in which corrections are applied. If a length-control step is fitted afterwards on the same single-order data, it absorbs the $\frac{1}{2}\imath_{PV}$ discrepancy into its length coefficient, so the error moves into that coefficient rather than disappearing. Part (a) is the operational point. Length control exists to make verbose-versus-terse comparisons fair, and that contrast is where an additive position correction fails, by an amount set by the interaction and by nothing the pipeline can tune.

Remark 1 (Why not log-odds). It is tempting to state the result for a logistic model $\operatorname{logit} \Pr(Y{=}1) = \gamma + \beta_P P + \beta_V V + \beta_{PV} PV$, with a marginal regression on P converging to $\beta_P + \frac{1}{2}\beta_{PV}$. That limit is the *linear* omitted-variable formula (Angrist & Pischke, 2009). Logistic coefficients are non-collapsible (Greenland et al., 1999), so the marginal slope differs from it, and it differs even when $\beta_{PV} = 0$. Proposition 1 avoids the problem by working on the scale that pipelines report, where averaging over v is linear. The same algebra holds verbatim for cell *log-odds* when a pipeline subtracts a constant on the logit scale. A pipeline that instead fits a marginal logistic regression on the position indicator incurs an additional non-collapsibility error, which is a separate failure mode from interaction.

Corollary 1 (Which pipelines are exposed)**.** *(i) Single-order judging with any additive position adjustment: exposed, by $\frac{1}{2}\imath_{PV}$ on the length contrast. (ii) Judging both orders and averaging on the reported scale: not exposed to $P{\times}V$. (Averaging log-odds and reporting a probability targets a slightly different estimand, by Jensen's inequality.) (iii) Stratified correction, with a position offset estimated separately within each length stratum: not exposed. It needs both-order calibration in each stratum, which is exactly the SSIT design. Interactions that do not involve position, such as length×identity, are untouched by any position correction.*

What an interaction of a given size costs. At the pooled estimate of §7, $\imath_{PV} \approx 0.047$, an additive correction distorts the verbose-versus-terse contrast by about 2.4 win-rate points. That distortion has a fixed sign, so it does not average away. For scale, the standard error of a single win rate near 50% on an 805-prompt benchmark of AlpacaEval's size is ≈ 1.8 points. For preference data, a shift of δ in the pick probability reverses the majority label of every comparison whose quality-only preference lies within δ of $\frac{1}{2}$ on the side the shift crosses. With a Beta$(2, 2)$ distribution of preferences this is about $1.5\,\delta$ of the affected comparisons, i.e. roughly 2% at $\delta = \frac{1}{4}\imath_{PV}$ and 4% at $\delta = \frac{1}{2}\imath_{PV}$. These labels are not random noise: they all favour the longer response.

4 The Swap-Symmetry Interaction Test

SSIT runs each of n calibration stimuli through all four (p, v) cells, with $m \geq 1$ independent judgments per cell, and computes, per stimulus,

$$\hat{d}_k = [\bar{Y}_k(1,1) - \bar{Y}_k(0,1)] - [\bar{Y}_k(1,0) - \bar{Y}_k(0,0)], \qquad \hat{\imath}_{PV} = \frac{1}{n}\sum_{k=1}^{n} \hat{d}_k, \tag{3}$$

where $\bar{Y}_k(p, v)$ is the mean coded outcome (1) in cell (p, v).

Proposition 2 (Design-based identification)**.** *If, for each stimulus, judgments in the four cells are independent draws with means $r_k(p, v)$, then $\mathbb{E}[\hat{d}_k \mid k] = \imath_k$, and $\hat{\imath}_{PV}$ is unbiased for the average of $\imath_k$ over the sampled stimuli, and for $\imath_{PV}$ when the stimuli are drawn from the population. No labels, no model of the judge, and no assumption about the per-stimulus quality gap are needed: the quality gap enters all four cells equally and cancels in each inner difference.*

The proof is linearity of expectation. The role of the design is that the *same* stimulus appears in all four cells, so $\hat{d}_k$ contrasts the judge with itself. A log-odds analogue, $\iota = [\operatorname{logit} r(1,1) - \operatorname{logit} r(0,1)] - [\operatorname{logit} r(1,0) - \operatorname{logit} r(0,0)]$, removes the quality gap only within a stimulus, and it is not estimable from cells near 0 or 1 with few judgments. We therefore treat log-odds quantities as secondary and estimate them with stimulus fixed effects (§5.2).

Inference. Stimuli are the independent units: repeated judgments of one stimulus, and duplicated stimuli, are not independent. We test $H_0 : \imath_{PV} = 0$ with the sign-flip randomization test on $\{\hat{d}_k\}$ (Fisher, 1935), enumerating all 2^n sign assignments when $n \leq 16$. It is valid when $\hat{d}_k$ is symmetric about zero under H_0, and approximately valid otherwise. Intervals use the wild cluster bootstrap with Rademacher weights (Cameron et al., 2008). With n clusters the smallest attainable two-sided p-value is 2^{1-n}, which for the pilot's $n = 12$ is 0.0005.

Robustness to the judge's decision rule (simulation). Proposition 2 implies that $\hat{\imath}_{PV}$ tracks the true $\imath_{PV}$ whatever the judge does. Table 1 checks this on six planted judges, 60 stimuli and 300 replications each. One judge has no link function: it adds Student-t noise to perceived quality and thresholds the result. The bias is at most 0.004 in every case.

How many stimuli. Since $\hat{\imath}_{PV}$ is a mean of n independent $\hat{d}_k$, a two-sided level-α test has power $\approx \Phi\big(|\imath_{PV}|\sqrt{n}/\sigma_d - z_{1-\alpha/2}\big)$ with $\sigma_d^2 = \operatorname{Var}(\hat{d}_k)$, so 80% power at $\alpha = 0.05$ needs

$$n_{80} \approx 7.85\,\sigma_d^2/\imath_{PV}^2. \tag{4}$$

σ_d combines between-stimulus heterogeneity with sampling noise, which shrinks as $1/m$. Both can be estimated from a pre-run of about ten stimuli. Table 2 gives simulated power for a judge of intermediate decisiveness. With the per-judge σ_d and $\hat{\imath}_{PV}$ observed in our pilot, (4) gives n_{80} between 30 and 52 distinct stimuli per judge, against the 12 we had. That gap is the main reason per-judge conclusions are out of reach in §7.

Table 1: SSIT recovers the true win-rate interaction under six judge decision rules (`sim_ssit_robustness.py`; 60 stimuli, 1 judgment per cell, 300 replications, no labels). "True $\imath_{PV}$" is computed by high-precision Monte Carlo for each planted judge. The smaller true value in the fourth row reflects a judge whose bias genuinely fades on easy stimuli; it is not attenuation of the estimator.

Planted judge	true $\imath_{PV}$	mean $\hat{\imath}_{PV}$ (sd)	bias	$\Pr(\hat{\imath}_{PV} > 0)$
additive logit plus interaction	0.244	0.241 (0.078)	−0.004	1.00
logit, wide quality-gap spread	0.204	0.201 (0.072)	−0.003	1.00
bias slopes vary with quality (symm.)	0.218	0.218 (0.083)	+0.001	1.00
bias fades when target clearly better	0.160	0.159 (0.078)	−0.001	0.98
probit link	0.331	0.329 (0.070)	−0.002	1.00
t-noise + threshold, no link function	0.234	0.230 (0.075)	−0.004	0.99

Table 2: Power of the SSIT wild-cluster test at the 5% level, 2 judgments per cell, a judge choosing the target about 80% of the time (`sim_ssit_power.py`, 500 replications). Columns count *distinct* stimuli.

true $\imath_{PV}$	$n = 20$	$n = 30$	$n = 50$	$n = 100$
0.05	0.08	0.11	0.11	0.16
0.10	0.14	0.20	0.26	0.47
0.15	0.25	0.35	0.52	0.76
0.20	0.38	0.54	0.75	0.95

5 Two ways to manufacture an interaction

5.1 Tie coding

Judge prompts commonly allow a tie, and analysis code commonly binarizes the verdict as "Response A chosen". That makes a tie count for slot B. Our own collection code did this, and so did the first analysis of the pilot. The convention looks harmless, but it is not swap-equivariant. A tie is unchanged when the responses swap slots, yet the coding assigns it to whichever response is in slot B.

Write $a(p, v)$ for the probability that the verdict names the target and $t(p, v)$ for the probability of a tie. Under slot coding, the target-chosen outcome has mean $a(1, v)$ when the target is in slot A and $a(0, v) + t(0, v)$ when it is in slot B, because a tie then counts for the target.

Proposition 3 (Tie-coding bias)**.** *Let $\imath_{PV}^{\text{slot}}$ be the interaction under slot coding and $\imath_{PV}^{\text{strict}}$ the interaction under the swap-equivariant coding that scores a tie as* 0*. Then*

$$\imath_{PV}^{\text{slot}} = \imath_{PV}^{\text{strict}} + \big[t(0,0) - t(0,1)\big], \qquad \Delta^{\text{slot}}(v) = \Delta^{\text{strict}}(v) - t(0, v). \tag{5}$$

The half coding (1) *gives $\imath_{PV}^{\text{half}} = \imath_{PV}^{\text{strict}} + \frac{1}{2}\,\imath_{PV}^{\mathsf{tie}}$, where $\imath_{PV}^{\mathsf{tie}}$ is the difference-in-differences of tie rates, which is itself swap-equivariant.*

Proof. Under slot coding the four cell means are $a(1, v)$ and $a(0, v) + t(0, v)$. Under strict coding they are $a(1, v)$ and $a(0, v)$. Take the difference-in-differences of each and subtract. The half coding adds $\frac{1}{2}t(p, v)$ to every cell. □

Corollary 2 (A position-invariant judge looks biased)**.** *If the judge's verdict distribution about the target does not depend on slot, so that $a(1, v) = a(0, v)$ and $t(1, v) = t(0, v) = t(v)$, then every swap-equivariant coding gives $\Delta(v) = 0$ and $\imath_{PV} = 0$. Slot coding instead reports a preference for slot B of size $t(v)$, and an interaction of size $t(0) - t(1)$.*

Lengthening one response usually makes the pair easier to separate, so ties fall when $v = 1$ and slot coding reports a *positive* interaction. In our pilot the bias term $t(0, 0) - t(0, 1)$ reaches +0.083 for one judge (Table 3). The pooled position main effect in log-odds moves from −0.61 under slot coding to +0.25 under half coding, which indicates that the apparent preference for slot B was largely tie coding. Dropping ties instead is

swap-equivariant, but it conditions on a post-treatment outcome whose rate depends on the cell. We therefore report the half coding as primary and strict coding as a sensitivity analysis.

5.2 Curvature near the ceiling

Suppose the judge's log-odds are additive, with no interaction at all:

$$\operatorname{logit} r_k(p, v) = \delta_k + \beta_P(2p - 1) + \beta_V v. \tag{6}$$

The win-rate interaction is still generally nonzero, because the slope of the logistic function depends on where a cell sits (Ai & Norton, 2003).

Proposition 4 (Curvature-only interaction)**.** *Under* (6)*,*

$$\imath^0_{PV} = \mathbb{E}_k\big[\sigma(\delta_k{+}\beta_V{+}\beta_P) - \sigma(\delta_k{+}\beta_V{-}\beta_P) - \sigma(\delta_k{+}\beta_P) + \sigma(\delta_k{-}\beta_P)\big] = 2\beta_P\,\mathbb{E}_k\big[\sigma'(\delta_k{+}\beta_V) - \sigma'(\delta_k)\big] + O(\beta_P^3).$$

In particular $\imath^0_{PV} = 0$ *whenever* $\beta_P = 0$*: curvature can bend a position effect but cannot create one.*

When the target already wins most of the time (δ_k large) and lengthening raises it further ($\beta_V > 0$), then $\sigma'(\delta_k + \beta_V) < \sigma'(\delta_k)$, and a *negative* β_P produces a *positive* $\imath^0_{PV}$. This is the configuration that slot-coded ties produce (§5.1), which is how the two artefacts compound. We estimate $\imath^0_{PV}$ for each judge by fitting (6) with stimulus fixed effects to the stimulus×cell table. Half-counts encode ties, and weak ridge penalties keep separated stimuli finite. We then plug the fit into the expression above, and assess the observed $\hat{\imath}_{PV}$ against a parametric bootstrap of the statistic under the fitted additive model. The same model with a $\beta_{PV}(2p-1)v$ term gives the log-odds interaction, with a stimulus-cluster bootstrap interval. The two estimands answer different questions. $\imath_{PV}$ is the operational one: does an additive win-rate correction leave a residual? It does, by Proposition 1, whatever the residual's source. The excess over $\imath^0_{PV}$ and β_{PV} are mechanistic: does the judge's latent position preference depend on length?

6 A label-free single-pass correction

When a pipeline can afford only one order per evaluated item and has no labels, the swap cells of a small calibration set still identify the position-coupled terms. **Interaction-Aware Swap Correction** (IASC) estimates $\hat{\beta}_P$ and $\hat{\beta}_{PV}$ from the pooled calibration cell log-odds by difference-in-differences, then corrects each single-order judgment: $\hat{s}_i = \sigma\big(\operatorname{logit}\hat{Y}_i - \hat{\beta}_P(2P_i - 1) - \hat{\beta}_{PV}(2P_i - 1)V_i\big)$.

Proposition 5 (IASC under a probit random-intercept judge)**.** *Let* $\Pr(Y = 1 \mid k, p, v) = \Phi\big(\delta_k + \beta_P(2p - 1) + \beta_V v + \beta_{PV}(2p - 1)v\big)$ *with* $\delta_k \sim \mathcal{N}(\mu, \tau^2)$*. The pooled cell rates then follow a probit model with every coefficient multiplied by* $\lambda = (1 + \tau^2)^{-1/2}$*. IASC on the probit scale, applied to pooled single-order rates, removes the position-coupled terms exactly. Applied to individual items, it removes the fraction* λ *of each position-coupled latent term, preserving its sign and leaving* β_V *untouched.*

The proof is the standard marginalization of a normal random intercept in a probit model; for the logit the scaling is approximate (Zeger et al., 1988). IASC targets a latent-scale estimand, while swap-averaging targets $\theta(v)$, and the two differ by a Jensen term. We therefore treat IASC as a floor, to be used only when a second pass is unaffordable. On the pilot (Appendix C), IASC moves the pooled single-order rate from 0.957 to 0.947, against 0.945 for full swap-averaging, and it moves in the direction of swap-averaging for every judge.

7 Pilot: seven judges, six families

Stimuli. The stimulus set has 12 hand-constructed templates, three in each of summarization, arithmetic word problems, explanatory question answering, and code review (Appendix A). Each template has a target r_1 and a comparison r_2. By construction r_2 is visibly weaker: it is shorter and less complete. r_1 is already longer than r_2 at v=0 (mean 47 vs. 24 words). The lengthened target comes in two forms. In the *paraphrase* form, r_1 is rewritten to 3–4× its length, which adds elaboration and, for code, some extra content and Markdown. In the *padding* form, r_1 is kept verbatim and three sentences restating its points are appended (about 1.9×

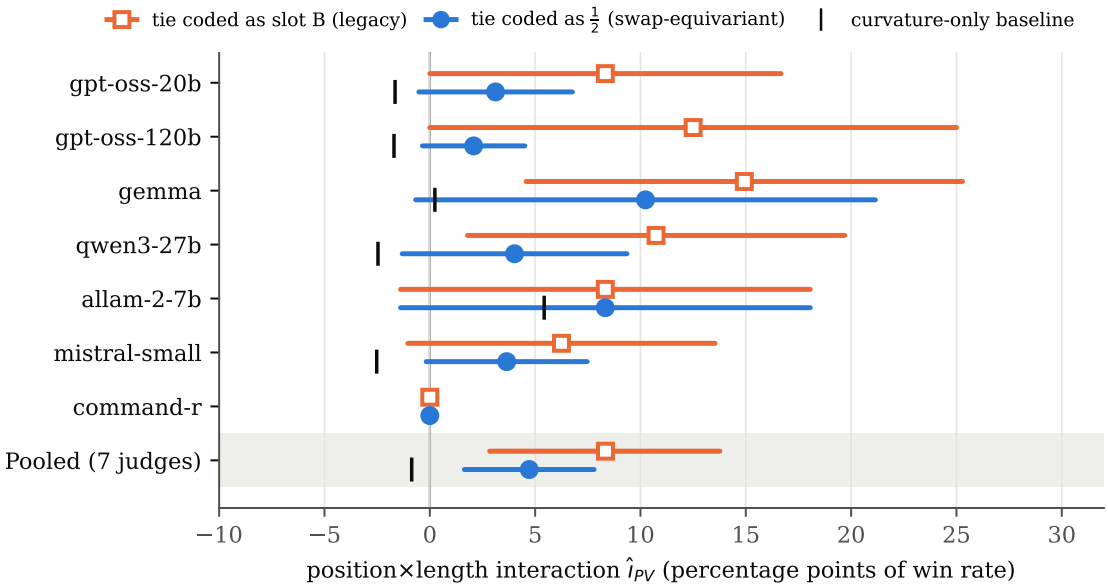


Figure 1: Position×length interaction per judge, paraphrase manipulation, 95% wild-cluster intervals over the 12 distinct stimuli. Squares: tie counted for slot B (slot coding). Circles: tie scored $\frac{1}{2}$ (swap-equivariant). Vertical ticks: the curvature-only interaction i^0_{PV} implied by an additive log-odds fit of the same judge under half coding. Slot coding inflates every judge that ties, and none of the swap-equivariant per-judge intervals is decisive. `allam-2-7b` never ties, so both codings coincide for it. The pooled interaction excludes zero and clears its curvature baseline.

its length). A third factor substitutes the judge's own generation for r_1 (self-preference, S). Every analysis below averages over S, which is balanced across cells. The collection script cycles through the templates, so the 18–20 "pairs" judged per judge are **12 distinct stimuli**. All inference clusters on these 12, treating duplicates as additional judgments of the same stimulus.

Judges and protocol. The judges are `gpt-oss-20b` and `gpt-oss-120b` (OpenAI, served by Groq), `gemma` (Google), `qwen3-27b` (Qwen), `allam-2-7b` (SDAIA), `mistral-small` (Mistral), and `command-r-08-2024` (Cohere). Six families are represented, and no frontier proprietary judge is included. The prompt asks which response is better "based solely on accuracy, relevance, and completeness" and instructs the judge *not* to factor in length, formatting, or order; a tie is allowed. Both `gpt-oss` judges gave one judgment per cell at temperature 0; the others gave two at temperature 0.7. In total there are 1,806 parsed judgments. Unparseable verdicts were dropped: 15% for `gemma`, 9% for `qwen3-27b`, 3% for `gpt-oss-20b`, and none for the rest (Appendix A). Data were collected in August 2026.

Per judge: nothing survives a careful analysis. Under slot coding the per-stimulus interaction is positive for six of seven judges (Figure 1). That is the pattern a naive analysis would report as broad evidence of interaction. Scoring ties as $\frac{1}{2}$ removes between 0 and 10 win-rate points per judge, in line with Proposition 3. After this correction, no judge's interval excludes zero by a margin that survives multiplicity correction, and with 12 stimuli no judge reaches nominal significance. The second artefact also shows up in the data. Under slot coding, `gpt-oss-120b`'s curvature baseline (+0.139) is *larger* than its observed interaction (+0.125): the additive model with the spurious slot-B effect reproduces the whole signal, so tie coding and curvature together account for this judge's apparent interaction. `command-r` picks the target on every judgment, which leaves SSIT nothing to detect. This is a ceiling property of the stimuli, and it says nothing about whether that judge is unbiased.

Table 3: SSIT on the pilot (paraphrase manipulation). $\hat{\imath}_{PV}$ in win rate (fraction); 95% wild-cluster CI and sign-flip p over 12 stimuli. "Ties": overall tie rate. $t_{00}-t_{01}$: the slot-coding bias term of Proposition 3. $\imath^0_{PV}$: curvature-only baseline. $\hat{\beta}_{PV}$: log-odds interaction from the stimulus-fixed-effects fit, with stimulus-cluster bootstrap CI. The last three columns use the half coding. No per-judge p-value survives Holm correction under either coding.

			slot coding		half coding (primary)			
Judge	ties	$t_{00}-t_{01}$	$\hat{\imath}_{PV}$	p	$\hat{\imath}_{PV}$ [95% CI]	p	$\imath^0_{PV}$	$\hat{\beta}_{PV}$ [95% CI]
`gpt-oss-20b`	.05	+.042	+.083	.25	+.031 [−.01, +.07]	.50	−.017	+1.17 [+0.08, +1.68]
`gpt-oss-120b`	.17	+.083	+.125	.25	+.021 [−.00, +.05]	.50	−.017	+0.40 [−0.00, +0.94]
`gemma`	.17	+.042	+.149	.06	+.102 [−.01, +.21]	.13	+.002	+0.63 [+0.20, +1.85]
`qwen3-27b`	.08	+.044	+.107	.13	+.040 [−.01, +.09]	.25	−.025	+1.51 [+0.50, +1.97]
`allam-2-7b`	.00	.000	+.083	.31	+.083 [−.01, +.18]	.31	+.054	+0.45 [−0.73, +0.90]
`mistral-small`	.10	−.021	+.062	.50	+.036 [−.00, +.07]	.25	−.025	+1.06 [+0.00, +1.93]
`command-r`	.01	.000	.000	1	.000	1	—	—
Pooled (7)	.07	+.020	+.083	.012	+**.047** [+**.02**, +**.08**]	**.012**	−.009	+**0.67** [+**0.35**, +**1.18**]
Pooled (excl. `command-r`)	.09	+.024	+.101	.012	+.058 [+.02, +.10]	.012	−.010	+0.69 [+0.39, +1.26]

Pooled: a modest interaction that survives every correction. Pooling the seven judges, with the 12 stimuli as clusters, gives $\hat{\imath}_{PV} = +0.047$ (95% CI $[0.016, 0.078]$, sign-flip $p = 0.012$). The curvature baseline is -0.009, so the excess is $+0.056$ (parametric bootstrap $p = 0.009$). The log-odds interaction is 0.67 $[0.35, 1.18]$, and strict coding gives $+0.063$ $[0.02, 0.11]$. Eight of the 12 stimuli have positive $\hat{d}_k$, three are exactly zero, and one is negative. The structure is the one Proposition 1 is about. When lengths match, the position effect is -0.1 points; when the target is lengthened, it is $+4.6$ points. Averaged over length, as a sequential pipeline estimates it, position bias is about $+2.3$ points. A pipeline that subtracts this average over-corrects matched-length comparisons and under-corrects lengthened ones, by about 2.3 points each. This also squares with reports that position bias is small in isolation (Soumik, 2026). A position effect that appears only alongside a length gap looks small on average and still breaks additive correction. These pooled statements treat the seven judges as fixed, so they generalize over stimuli, not over judges.

Length or rewriting? The paraphrase manipulation entangles length with rewriting. Four judges were re-collected with the padding manipulation, which changes length alone. On those same four judges, the paraphrase interaction is $+0.038$ $[0.000, 0.077]$ and the padding interaction is $+0.027$ $[-0.004, 0.057]$ ($p = 0.13$). Padding thus keeps about 70% of the magnitude, but on four judges and 12 stimuli it is not significant, and its log-odds interaction is 0.32 $[0.03, 0.88]$. Per judge, the padding estimates are $+0.052$ (`allam-2-7b`), $+0.036$ (`mistral-small`), $+0.027$ (`qwen3-27b`), and 0 (`command-r`), all with $p \geq 0.25$. We read this as consistent with a position×length component, without establishing one. Per-judge contrasts between the two manipulations, such as an interaction that "vanishes" for one judge under padding, do not survive tie correction and stimulus clustering, and we draw no conclusions about mechanism from them.

Other factor pairs and tasks. Under half coding, position×self-preference is $+0.034$ $[-0.01, 0.07]$ ($p = 0.15$) and length×self-preference is -0.040 $[-0.10, 0.02]$ ($p = 0.25$). Neither is distinguishable from zero. By task (three stimuli each, no inference), the interaction is largest for arithmetic ($+0.087$) and smallest for open QA ($+0.016$), and positive in all four.

8 Discussion

A procedure. For any evaluation whose position correction is additive: (1) Code outcomes swap-equivariantly, with ties scored $\frac{1}{2}$, and report strict coding as a sensitivity check. (2) Run SSIT on n *distinct* stimuli, with n from (4) using σ_d from a ten-stimulus pre-run; the pilot suggests $n \approx 30$–50 per judge. (3) Report $\hat{\imath}_{PV}$ with a wild-cluster interval and sign-flip p, together with the curvature baseline $\imath^0_{PV}$ and $\hat{\beta}_{PV}$. (4) If the interval excludes zero, replace the additive correction with swap-averaging or stratified correction

(Corollary 1). If the budget allows only one order and there are no labels, use IASC as a floor. If the interval includes zero but (4) at the observed σ_d gives power below 0.8, the result is inconclusive and should not be read as additivity. For disclosure, an evaluation card for a judge-based pipeline should state how position is corrected, the outcome coding of ties, and $\hat{\imath}_{PV}$ with its uncertainty.

Relation to label-based methods. SSIT asks whether a structural assumption of the debiasing pipeline holds, and it needs no labels. Prediction-powered inference (Angelopoulos et al., 2023a;b) attaches valid intervals to a debiased metric using a gold set. If both are available, the gold-set rectifier should be stratified by (p, v) cell. Otherwise an interaction shifts the rectifier between calibration and evaluation whenever the cell mix differs.

Limitations. The pilot is small. It has 12 distinct, hand-constructed stimuli; a visibly weaker comparison response, which places judges near the ceiling where both power and the curvature argument are least favourable; one or two judgments per cell; no frontier proprietary judges; and padding data for only four judges. The padding sentences were generated by `mistral-small`, which is also a judge. The paraphrase manipulation changes content and formatting as well as length. The pooled result therefore establishes that SSIT detects a position×length interaction across these judges and stimuli. It does not establish the size of the effect for any particular deployed judge. IASC was developed after the data were collected. The sign-flip test assumes $\hat{d}_k$ is symmetric under the null, and with 12 clusters the wild bootstrap is coarse. The confirmatory protocol (Appendix D) addresses each of these limitations.

Conclusion. Whether judge biases compose is a property of the whole pipeline, not of any single bias, and it can be checked with the judge alone. The check is cheap in labels, about eight times the cost of one judgment per stimulus for a calibration subset, but it is not free of pitfalls. Two analysis choices that look innocuous, mapping ties to a slot and reading win-rate interaction as mechanism, can each produce the effect being measured. Done carefully, the check still finds a modest position×length interaction across seven judges, of the kind that additive corrections leave behind.

Broader Impact Statement

Judge-based scores determine leaderboard positions and preference data for training. An interaction left uncorrected biases both in a fixed direction, here toward longer responses, so a cheap diagnostic has practical value. The opposite risk is equally real. Our own first analysis overstated the effect through a coding convention, and published per-model bias magnitudes from small studies can be mistaken for properties of named models. Pilot magnitudes in this paper should not be cited as measurements of any deployed judge.

Reproducibility Statement

All pilot numbers, Table 3, Figure 1 and Appendices A–C are produced by `analyze_tmlr.py` and `make_tmlr_figures.py` from the released judgment files, with seed 2024, in about five minutes on a laptop. Tables 1–2 are produced by `sim_ssit_robustness.py` and `sim_ssit_power.py`. The collector (`collect_free.py`) records every raw verdict, so alternative codings can be applied without re-querying. Code and data: anonymized for review.

References

Chunrong Ai and Edward C. Norton. Interaction terms in logit and probit models. *Economics Letters*, 80(1): 123–129, 2003.

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023a.

Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*, 2023b.

Joshua D. Angrist and Jörn-Steffen Pischke. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, 2009.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008.

Y. T. Chen, S. Lu, S. Li, M. Guo, and S. Li. Efficient inference for noisy LLM-as-a-judge evaluation. *arXiv preprint arXiv:2601.05420*, 2026.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *International Conference on Machine Learning*, 2024.

Roger R. Davidson. On extending the Bradley-Terry model to accommodate ties in paired comparison experiments. *Journal of the American Statistical Association*, 65(329):317–328, 1970.

Yann Dubois, Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. AlpacaFarm: A simulation framework for methods that learn from human feedback. *Advances in Neural Information Processing Systems*, 2023.

Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. In *Conference on Language Modeling*, 2024.

Ronald A. Fisher. *The Design of Experiments*. Oliver and Boyd, 1935.

Sander Greenland, Judea Pearl, and James M. Robins. Confounding and collapsibility in causal inference. *Statistical Science*, 14(1):29–46, 1999.

Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Yuanzhuo Wang, and Jian Guo. A survey on LLM-as-a-judge. *arXiv preprint arXiv:2411.15594*, 2024.

Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. Benchmarking cognitive biases in large language models as evaluators. *arXiv preprint arXiv:2309.17012*, 2023.

Harrison Lee, Samrat Phatale, Hassan Mansoor, Kellie Lu, Thomas Mesnard, Colton Bishop, Victor Carbune, and Abhinav Rastogi. RLAIF: Scaling reinforcement learning from human feedback with AI feedback. *arXiv preprint arXiv:2309.00267*, 2023.

Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhattacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. From generation to judgment: Opportunities and challenges of LLM-as-a-judge. *arXiv preprint arXiv:2411.16594*, 2024.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, 2024.

P. V. Rao and L. L. Kupper. Ties in paired-comparison experiments: A generalization of the Bradley-Terry model. *Journal of the American Statistical Association*, 62(317):194–204, 1967.

Keita Saito, Akifumi Wachi, Koki Wataoka, and Youhei Akimoto. Verbosity bias in preference labeling by large language models. *arXiv preprint arXiv:2310.10076*, 2023.

Lin Shi, Chiyu Ma, Wenhua Liang, Weicheng Ma, and Soroush Vosoughi. Judging the judges: A systematic study of position bias in LLM-as-a-judge. *arXiv preprint arXiv:2406.07791*, 2024.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. A long way to go: Investigating length correlations in RLHF. *arXiv preprint arXiv:2310.03716*, 2023.

S. K. Soumik. Judging the judges: A systematic evaluation of bias mitigation strategies in LLM-as-a-judge pipelines. *Transactions on Machine Learning Research*, 2026. arXiv:2604.23178.

Aman Singh Thakur, Kartik Choudhary, Venkat Srinik Ramayapally, Sankaran Vaidyanathan, and Dieuwke Hupkes. Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-judges. *arXiv preprint arXiv:2406.12624*, 2024.

Tyler J. VanderWeele and Mirjam J. Knol. A tutorial on interaction. *Epidemiologic Methods*, 3(1):33–72, 2014.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2024.

Koki Wataoka, Tsubasa Takahashi, and Ryokan Ri. Self-preference bias in LLM-as-a-judge. *arXiv preprint arXiv:2410.21819*, 2024.

Minghao Wu and Alham Fikri Aji. Style over substance: Evaluation biases for large language models. *arXiv preprint arXiv:2307.03025*, 2023.

Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. Justice or prejudice? quantifying biases in LLM-as-a-judge. In *International Conference on Learning Representations*, 2025.

Scott L. Zeger, Kung-Yee Liang, and Paul S. Albert. Models for longitudinal data: A generalized estimating equation approach. *Biometrics*, 44(4):1049–1060, 1988.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2023.

A Stimuli, judges, and data

Templates. There are three templates per task. Summarization uses three expository passages, with target summaries of 2–3 sentences. Arithmetic uses three multi-step word problems (cost totals, area/perimeter, speed/time). Explanatory QA uses three "explain the difference/why" questions. Code review uses three short Python functions with a bug or inefficiency. Mean lengths in words are as follows. Target r_1: 41, 35, 54, 60 for the four tasks in order. Comparison r_2: 27, 16, 31, 21. Paraphrase-lengthened r_1: 135, 108, 227, 199. Padding-lengthened r_1: 85 against 45 for r_1, over the stimuli used. One template in full:

> **Prompt.** Summarize the following passage in 2–3 sentences: *The Amazon rainforest, often referred to as the 'lungs of the Earth,' covers over 5.5 million square kilometers ...*
> r_1**.** The Amazon rainforest spans over 5.5 million square kilometers and hosts roughly 10% of Earth's species, including tens of thousands of plants and animals. It is critically important to global ecology but has lost approximately 17% of its area over the last 50 years due to agriculture, logging, and development.
> r_2**.** The Amazon is a big rainforest in South America. It has lots of animals and plants. Some of it has been cut down for farming and other uses.

Judges.

Label	Model identifier	Provider	Family	judgm./cell	temp.	parsed
`gpt-oss-20b`	`openai/gpt-oss-20b`	Groq	OpenAI	1	0	140/144
`gpt-oss-120b`	`openai/gpt-oss-120b`	Groq	OpenAI	1	0	144/144
`gemma`	Gemma (see note)	Google	Gemma	2	0.7	271/320
`qwen3-27b`	`qwen/qwen3.8-27b`	Groq	Qwen	2	0.7	291/320
`allam-2-7b`	`allam-2-7b`	Groq	ALLaM	2	0.7	320/320
`mistral-small`	`mistral-small-latest`	Mistral	Mistral	2	0.7	320/320
`command-r`	`command-r-08-2024`	Cohere	Cohere	2	0.7	320/320

Unparseable responses were mostly deliberations that stopped without a verdict line. They were more frequent in lengthened cells. Under padding, `qwen3-27b`'s drop rate rose to 16% (268/320 parsed). Dropping is per judgment; a stimulus with an empty cell for a judge would leave that judge's analysis, but no such case occurred.

B Proofs

Proposition 4. Under (6) the cell rates are $\sigma(\delta_k + \beta_V v \pm \beta_P)$, so the per-stimulus difference-in-differences equals $g(\delta_k + \beta_V) - g(\delta_k)$ with $g(x) = \sigma(x + \beta_P) - \sigma(x - \beta_P)$. Then $g \equiv 0$ when $\beta_P = 0$, and Taylor expansion gives $g(x) = 2\beta_P \sigma'(x) + O(\beta_P^3)$.

Proposition 5. Write $Y = \mathbb{1}\{\delta_k + x^\top \beta + \varepsilon > 0\}$ with $\varepsilon \sim \mathcal{N}(0,1)$ independent of $\delta_k \sim \mathcal{N}(\mu, \tau^2)$. Marginally, $\delta_k + \varepsilon \sim \mathcal{N}(\mu, 1 + \tau^2)$, so $\Pr(Y = 1 \mid x) = \Phi\big((\mu + x^\top \beta)/\sqrt{1 + \tau^2}\big)$. All pooled-cell coefficients are therefore scaled by the same λ, and the swap-cell difference-in-differences of pooled probits returns $\lambda\beta_P$ and $\lambda\beta_{PV}$. Subtracting these from the pooled single-order probit leaves $\lambda(\mu + \beta_V v)$, which is position-free. Subtracting them from an item's latent index $\delta_k + x^\top \beta$ leaves $(1 - \lambda)$ of each position-coupled term.

Exactness of the identity in Proposition 3. `analyze_tmlr.py` recomputes both sides per stimulus for every judge and both manipulations; the maximum absolute discrepancy is 1.1×10^{-16}.

C IASC on the pilot

Judge	$\hat{\beta}_P$	$\hat{\beta}_{PV}$	raw (one order)	+IASC	swap-avg. (2×)
`gpt-oss-20b`	−0.03	+0.77	0.984	0.983	0.977
`gpt-oss-120b`	+0.06	+0.31	0.948	0.941	0.938
`gemma`	+0.24	+0.52	0.945	0.925	0.890
`qwen3-27b`	+0.10	+1.00	0.977	0.973	0.962
`allam-2-7b`	−0.87	+0.46	0.875	0.923	0.917
`mistral-small`	+0.38	+0.77	0.977	0.963	0.944
`command-r`	+0.35	0.00	1.000	0.999	0.997
Pooled	−0.01	+0.62	0.957	0.947	0.945

Half coding throughout; coefficients are estimated in-sample from the pooled cell log-odds. "Raw" is the target-chosen rate when the target is shown first. IASC moves every judge toward the swap-averaged rate. It overshoots slightly only for `allam-2-7b` (0.923 vs. 0.917), and recovers 83% of the raw-to-swap gap when pooled. This is a consistency check, not a validation: the pilot has no ground truth.

D Protocol for a confirmatory study

- **Judges.** At least eight, from at least five families, including frontier proprietary judges, and fixed before collection.
- **Stimuli.** At least 50 *distinct* stimuli per judge (from (4) with pilot σ_d), drawn from public benchmarks across four or more task types. Pairs are quality-matched by an independent rater so that slot-balanced win rates lie near $\frac{1}{2}$. This maximizes power and minimizes curvature.
- **Manipulation.** Length-only padding as primary, generated by a model outside the judge panel, with paraphrase lengthening secondary. Length is lengthened in both directions (target and comparison) on separate stimuli.
- **Collection.** All (p, v) cells, cell order randomized per stimulus, at least 3 judgments per cell at fixed temperature, and raw verdict text logged.

- **Primary endpoint.** $\hat{\imath}_{PV}$ under half coding, per judge and pooled. Sign-flip test over stimuli, Holm-adjusted over judges. Decision rule: the interaction is declared present if the adjusted $p < 0.05$ and $|\hat{\imath}_{PV}| > 0.02$.
- **Secondary endpoints.** Strict coding; excess over the curvature baseline; $\hat{\beta}_{PV}$; position effect at each length level; the padding-to-paraphrase ratio.
- **Pre-registration.** Endpoint, decision rule, coding, and sample size registered before collection. IASC evaluated on held-out stimuli.